Full-Trajectory Reconstruction for UHR-SAR Large-Scene Imaging via Joint Residual Motion Errors and Dominant Target Positions Estimation

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***Abstract*—Advances in imaging applications require radars to achieve higher resolution and cover larger scenes. Accurate full-trajectory information is crucial for obtaining well-focused large-scene ultrahigh-resolution (UHR) synthetic aperture radar (SAR) images with centimeter-level resolution. However, the positioning accuracy of most inertial navigation systems (INS)/global positioning system (GPS), even high-end models such as the POS AV 610 RTX, is inadequate to meet such imaging requirements, thus necessitating full-trajectory reconstruction. This process involves two major challenges. First, in UHR imaging, residual motion errors induce nonsystematic range cell migration (RCM) that exceeds half a range cell, causing severe 2-D defocusing. This defocusing not only hinders accurate dominant target position determination but also complicates the estimation of azimuth-phase errors. Second, in large-scene imaging, the full-trajectory spans multiple full apertures, but the information provided by any dominant target is confined to one aperture, making direct full-trajectory reconstruction impossible. To overcome these challenges, this article improves the conventional range history model through a first-order Taylor expansion with respect to the residual motion errors and dominant target position errors, and formulates a range history error minimization problem. To efficiently solve the resulting complex high-dimensional optimization, a dimensionality reduction strategy is developed, which enables joint estimation of the residual motion errors and the dominant target positions. Combined with trajectory fusion, the proposed method reconstructs a full-trajectory capable of covering the entire large scene. Simulations and experimental results demonstrate that the reconstructed trajectory provides sufficient accuracy to support well-focused UHR large-scene imaging.**

***Index Terms*—Dominant target position errors, residual motion errors, trajectory fusion, trajectory reconstruction, ultrahigh-resolution (UHR) large-scene imaging.**

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I. Introduction

ULTRAHIGH-RESOLUTION (UHR) large-scene imaging can capture rich spatial information, enabling more detailed target characterization and improving the accuracy of target recognition [1], [2], [3], [4], [5]. Microwave photonic synthetic aperture radar (MWP-SAR) is one of the promising approaches for UHR imaging [6], [7], [8], [9], capable of achieving centimeter- or even millimeter-level resolution [3], [10], [11], [12], featuring lightweight design, low power consumption, and resistance to electromagnetic interference, thus being widely deployed on different platforms [13], [14], [15], [16], [17], [18], [19].

The key to obtaining a well-focused large-scene UHR-SAR image lies in the availability of an accurate full trajectory, regardless of whether time-domain imaging algorithms (TDAs) or frequency-domain imaging algorithms (FDAs) are employed, as also recognized in early airborne SAR studies with 10-cm resolution [20]. Specifically, for TDAs such as the backprojection algorithm (BPA), an accurate trajectory alone suffices to achieve UHR large-scene focusing [21]. For FDAs, however, including the range-Doppler algorithm (RDA) and the range migration algorithm (RMA), the nonlinearities in platform motion induce 2-D spatial-variant motion errors that adversely affect both the echo envelope and phase, making compensation essential [2], [22], [23]. To address these errors, many methods have been proposed. For example, the direct motion algorithm (DMA) [24] and methods combining the two-step algorithm (TSA) with chirp-Z transform (CZT) [25], [26], which utilize the measured trajectory information, have been developed to correct range spatial-variant motion errors. In addition, a variety of techniques reliant on measured trajectory data have been introduced to address azimuth-variant errors [27]. These include subaperture methods, such as the precise topography and aperture (PTA)-dependent [28],

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subaperture topography and aperture (SATA)-dependent [29], and frequency-division (FD) [30] algorithms, as well as full-aperture approaches based on nonlinear chirp scaling (NCS) [31] or resampling (RS) [32]. When measured trajectory data are accurate, these compensation techniques can effectively correct the 2-D spatial-variant motion errors, thereby enabling FDAs to achieve UHR large-scene focusing as well.

In practice, however, whether for TDAs like BPA or for the aforementioned compensation techniques applied to FDAs, such methods often prove ineffective. This is the case when the positioning accuracy of the inertial navigation systems (INSs) and global positioning systems (GPSs) is insufficient to meet the stringent trajectory precision required by UHR imaging. Consequently, motion error compensation methods that operate directly on radar raw data become necessary. Among them, several well-established algorithms such as the phase gradient algorithm (PGA) [33], the MapDrift algorithm (MDA) [34], and BP autofocus [35], [36] are mature, computationally efficient, and robust. However, they cannot handle the severely 2-D spatial-variant motion errors encountered in UHR large-scene imaging. Recently, several novel methods capable of compensating 2-D spatial-variant motion errors have been proposed and have shown promising focusing performance in UHR large-scene imaging [10], [27], [35], [36], [37], [38].

In parallel, trajectory reconstruction provides another approach to address residual-motion-error-induced defocusing in UHR large-scene imaging. Within the last few years, considerable efforts have been devoted to trajectory reconstruction, and a variety of approaches have been proposed. Hai et al. [16] divide the full aperture into multiple subapertures and estimate the linear trajectory errors of adjacent subapertures through image registration, which essentially follows the polynomial fitting strategy. In the case of UHR imaging, the requirement for higher reconstruction accuracy becomes more stringent, for which nonpolynomial reconstruction provides a more suitable framework. Within this framework, Pu et al. [39] and Zhong et al. [40] iteratively optimize the residual motion errors over all azimuth time samples by minimizing image entropy (IE), thereby achieving nonpolynomial trajectory reconstruction that meets the demands of UHR imaging. However, these methods suffer from high computational complexity and low efficiency, which leads to a prohibitive computational burden when applied to large-scene imaging. To alleviate this, Zhang et al. [38] and Liang et al. [41] proposed dominant target-based trajectory reconstruction, where residual motion errors are calculated via least squares (LS) without iterative optimization, offering low complexity and high efficiency. Nevertheless, its performance relies heavily on azimuth-phase error estimation results and dominant target position, which are difficult to obtain from UHR coarse imaging results. Moreover, this method can only reconstruct a single full-aperture-length trajectory, making it insufficient for large-scene imaging. In summary, efficient reconstruction of a full-trajectory for UHR large-scene focusing still faces two critical challenges.

1) For UHR imaging, the large transmitted bandwidth yields extremely small range cells. Consequently, the nonsystematic range cell migration (NsRCM) induced by residual motion errors easily exceeds half a range cell, resulting in severe 2-D defocusing in coarse imaging. This severely degrades dominant target position determination and complicates azimuth-phase error estimation, making trajectory reconstruction infeasible directly from coarse imaging results.
2) For large-scene imaging, the full-trajectory spans multiple full apertures. However, each dominant target can only provide at most one full aperture of information and thereby supports only a single full-aperture-length trajectory reconstruction. This limitation makes direct full-trajectory reconstruction infeasible.

With the continuing increase in imaging requirements, the transmitted signal bandwidth is expected to further increase and may even exceed 10 GHz [10], [12], [42], and the imaged scene is also expected to expand. Both trends will further exacerbate the two challenges encountered in UHR large-scene imaging. In this article, we propose a full-trajectory reconstruction method to overcome these challenges. First, considering the difficulty in obtaining dominant target positions and azimuth-phase errors in challenge 1), a stepwise reconstruction strategy is adopted, in which trajectory reconstruction is divided into range-envelope-level reconstruction and azimuth-phase-level reconstruction. The purpose of the former is to ensure that the NsRCM induced by residual motion errors does not exceed half a range cell. This step reduces image defocusing from 2-D to 1-D, thereby facilitating the estimation of azimuth-phase errors. At the same time, since dominant target positions are inaccurate, the range history model is improved by applying a first-order Taylor expansion with respect to the dominant target position and the residual motion error. This article formulates a range history error minimization problem to estimate these errors. The position errors are treated as key unknown parameters to be estimated jointly with the residual motion errors. A dimensionality reduction strategy is further employed to improve estimation efficiency, that is, the optimal trajectory reconstruction can be achieved by optimizing only the dominant target positions. Then, to obtain a full-trajectory covering multiple full apertures, multiple independently reconstructed overlapping subtrajectories are optimally fused based on the minimum waveform entropy criterion, thereby addressing challenge 2). Airborne MWP-SAR data are used in this article to validate the effectiveness of the proposed method.

The remainder of this article is organized as follows. Section II establishes the echo model and analyzes the tolerances of residual motion errors and dominant target position errors at the range envelope level and azimuth phase level. Section III introduces the proposed full-trajectory reconstruction method. Simulation and measured results are presented in Section IV. Finally, Section V concludes this article.

II. Echo Model and Error Tolerance Analysis

In this section, Section II-A establishes the echo model obtained via deramping. Based on this model, Section II-B analyzes the residual motion error tolerances at both the range envelope level and the azimuth phase level, while Section II-C analyzes the dominant target position error tolerances also at both levels. The former analysis illustrates the inadequacy of

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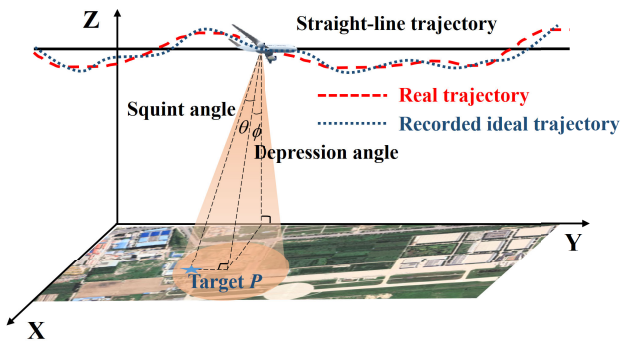


Fig. 1. Configuration of MWP-SAR system.

existing INS/GPS positioning accuracy and explains why 2-D defocusing occurs in the image. The latter analysis clarifies why inaccuracies in dominant target positions affect trajectory reconstruction and highlights the limitations of coarse image.

A. Echo Model With Residual Motion Error

In UHR imaging, deramping is commonly employed to reduce the sampling rate and alleviate data load. The airborne MWP-SAR system mentioned in this article adopts the same approach, with its configuration shown in Fig. 1. The black solid line, red dashed line, and blue dotted line represent the straight-line, real, and recorded ideal trajectory, respectively. With a linearly frequency-modulated (LFM) transmitted signal, the received echo from an arbitrary target P before deramping is expressed as

$$s_r(\tau,\eta) = \text{rect}\left(\frac{\tau - t_r(\eta)}{T_r}\right)\text{rect}\left(\frac{\eta}{T_a}\right) \exp\left\{j\left[K_r\pi(\tau - t_r(\eta))^2 + 2\pi f_c(\tau - t_r(\eta))\right]\right\} \quad (1)$$

where τ and η denote the fast time and slow time, respectively; T_r, T_a, K_r, and f_c represent the pulse duration, synthetic aperture time, transmitted chirp rate, and carrier frequency, respectively; $t_r(\eta)$ represents the real delay of target P and $t_r(\eta) = 2R_r(\eta)/c$; c is the speed of light; and $R_r(\eta)$ is the real range history of target P, expressed as

$$R_r(\eta) = \sqrt{(x_r(\eta) - x_p)^2 + (y_r(\eta) - y_p)^2 + (z_r(\eta) - z_p)^2} \quad (2)$$

where $(x_r(\eta), y_r(\eta), z_r(\eta))$ and (x_p, y_p, z_p) denote the coordinate of real trajectory and target P, respectively.

The real trajectory $(x_r(\eta), y_r(\eta), z_r(\eta))$ can be expressed as the ideal trajectory $(x_i(\eta), y_i(\eta), z_i(\eta))$ pulse residual motion errors $(\Delta x(\eta), \Delta y(\eta), \Delta z(\eta))$, where the ideal trajectory refers to the trajectory recorded by the INS/GPS. By performing a first-order Taylor expansion of the real range history with respect to residual motion errors and evaluating it at zero error, the relationship between the ideal and real range histories is obtained

$$\begin{aligned} R_r(\eta) &= R_i(\eta) + \Delta R(\eta) \\ &\approx R_i(\eta) + \left(\frac{x_i(\eta) - x_p}{R_i(\eta)}\right)\Delta x(\eta) \\ &\quad + \left(\frac{y_i(\eta) - y_p}{R_i(\eta)}\right)\Delta y(\eta) + \left(\frac{z_i(\eta) - z_p}{R_i(\eta)}\right)\Delta z(\eta) \end{aligned} \quad (3)$$

where $\Delta R(\eta)$ denotes the range history error and $R_i(\eta)$ denotes the ideal range history, expressed as follows:

$$R_i(\eta) = \sqrt{(x_i(\eta) - x_p)^2 + (y_i(\eta) - y_p)^2 + (z_i(\eta) - z_p)^2}. \quad (4)$$

In the process of deramping, a reference LFM signal is used for mixing instead of a single-frequency signal. It should be noted that the reference delay of the reference signal in the MWP-SAR system mentioned in this article is zero, and the expression is as follows:

$$h(\tau,\eta) = \text{rect}\left(\frac{\tau}{T_r}\right)\text{rect}\left(\frac{\eta}{T_a}\right)\exp\left[j\left(K_r\pi\tau^2 + 2\pi f_c\tau\right)\right]. \quad (5)$$

By multiplying (1) and the conjugates of (5), the signal after deramping can be obtained. Then, pulse compression is performed via a Fourier transform (FT) to the fast time, and the expression ignoring amplitude information is given as follows:

$$s_{r_deramp}(f_\tau,\eta) = \text{sinc}\left[T_r(f_\tau + K_r t_r(\eta))\right]\text{rect}\left(\frac{\eta}{T_a}\right) \exp\left[j\left(K_r\pi {t_r}^2(\eta) - 2\pi f_c t_r(\eta)\right)\right] \quad (6)$$

where f_τ denotes the range frequency and $K_r\pi {t_r}^2(\eta)$ is the residual video phase (RVP).

In the process of obtaining (6) using the FT, $f_\tau = -K_r t_r$ can also be derived. This relationship indicates that the physical meaning of range frequency is time delay or distance. Based on this, a variable substitution on f_τ in (6) is performed after removing the RVP, expressed as

$$s_{r_deramp}(\tau',\eta) = \text{sinc}\left[T_r K_r(\tau' - t_r(\eta))\right]\text{rect}\left(\frac{\eta}{T_a}\right)\exp(-j2\pi f_c t_r(\eta)) \quad (7)$$

where τ' denotes the new fast time, $\tau' \in [-f_s/2K_r, f_s/2K_r]$, and f_s is the range sample rate. The sample time of new fast time can be obtained by the following equation:

$$\Delta\tau' = \frac{f_s}{K_r N_\tau} = \frac{f_s}{K_r T_r f_s} = \frac{1}{B_\tau} \quad (8)$$

where N_τ is the number of range sampling points and B_τ is the bandwidth of transmit signal.

To facilitate the analysis of the residual motion error and dominant target position error tolerances at range envelope level in Sections II-B and II-C, the new fast time τ' and real delay $t_r(\eta)$ in (7) is expressed in terms of the new range r_τ and range history, respectively, yielding the following echo expression:

$$s_{r_deramp}(r_\tau,\eta) = \text{sinc}\left[\frac{2B_\tau}{c}(r_\tau - R_r(\eta))\right]\text{rect}\left(\frac{\eta}{T_a}\right)\exp\left(-j\frac{4\pi f_c}{c}R_r(\eta)\right). \quad (9)$$

From (3), (8), and (9), two pieces of information can be obtained. On the one hand, the size of new range cell is $c/(2B_r)$, which is the same as the range resolution. On the other hand, residual motion errors introduce range history errors, which in turn affect imaging results through both the range envelope and the azimuth phase. In this article, the offset of the range envelope caused by residual motion errors is referred to as the NsRCM induced by residual motion errors.

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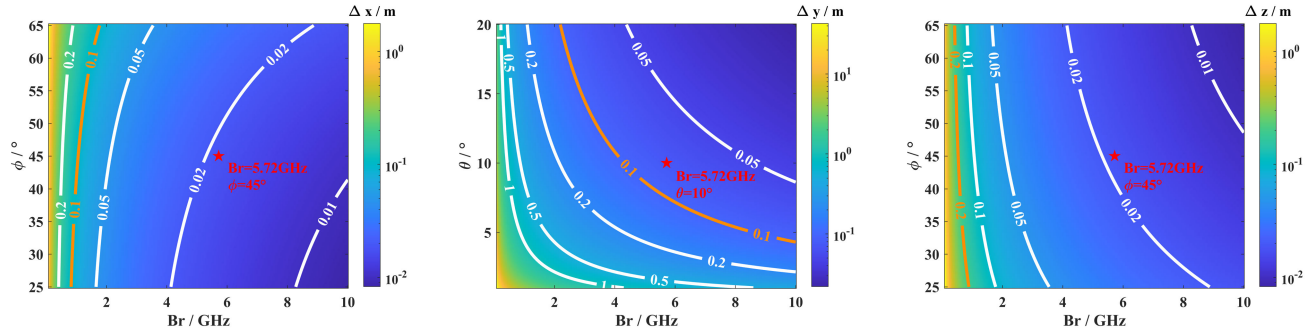


Fig. 2. Platform positioning accuracy required to prevent the RCM offset from exceeding half of a range cell. (a) Across-track direction, (b) along-track direction, and (c) vertical direction.

B. Residual Motion Error Tolerance Analysis

As mentioned above, residual motion errors result in NsRCM and azimuth-phase errors. In the following, we first analyze their impact on NsRCM and derive the corresponding tolerance at the range envelope level.

Since UHR imaging transmits signals with much larger bandwidth, the size of range cell becomes very small. Given the same platform positioning accuracy, NsRCM in UHR imaging is more likely to exceed half a range cell compared with conventional radar systems. This prevents TDAs from accurately tracing the RCM of targets, and FDAs combined with motion compensation cannot accomplish RCM correction (RCMC), further resulting in severe 2-D defocusing in the imaging results. Therefore, at the range-envelope level, the requirement for NsRCM is that it does not exceed half a range cell. That is, the range history error is required to satisfy the following expression:

$$|\Delta R(\eta)| < \frac{c}{4B_r}. \tag{10}$$

By substituting (3) into (10), the horizontal and vertical positioning accuracy requirements of the platform can be obtained, which correspond to the tolerances on the residual motion errors at the range envelope level

$$\begin{cases} |\Delta x(\eta)| < \dfrac{c}{4B_r}\left|\dfrac{R_i(\eta)}{x_i(\eta)-x_p}\right| = \dfrac{c}{4B_r\cos\phi\cos\theta} \\ |\Delta y(\eta)| < \dfrac{c}{4B_r}\left|\dfrac{R_i(\eta)}{y_i(\eta)-y_p}\right| = \dfrac{c}{4B_r\sin\theta} \\ |\Delta z(\eta)| < \dfrac{c}{4B_r}\left|\dfrac{R_i(\eta)}{z_i(\eta)-z_p}\right| = \dfrac{c}{4B_r\sin\phi\cos\theta} \end{cases} \tag{11}$$

where ϕ and θ denote the depression angle of antenna and squint angle of target P, respectively, shown in Fig. 1.

In practice, $\Delta x(\eta)$, $\Delta y(\eta)$, and $\Delta z(\eta)$ are coupled and jointly contribute to the range envelope, resulting in NsRCM. Here, this article makes a simplification by considering the positioning accuracy requirements in each direction independently. This is reasonable because what is derived represents the minimum positioning accuracy requirement for each direction.

From (11), it can be seen that the signal bandwidth, the target squint angle, and the depression angle determine the requirement on platform positioning accuracy in the along-track, across-track, and vertical directions.

TABLE I
APPLANIX POS AV ABSOLUTE ACCURACY SPECIFICATIONS

POS/AV 610 RTX	Post-processed
Position (m)	< 0.1H < 0.2V
Velocity (m/s)	0.030
True Heading (deg)	0.020
Roll / Pitch	0.005

Specifically, a larger signal bandwidth leads to higher positioning accuracy requirements in all three directions. In stripmap mode, a wider azimuth beamwidth enlarges the radar azimuth illumination angle. This results in a broader squint angle range, which also raises the accuracy requirements in all three directions. A wider range beamwidth enlarges the depression angle range, increasing the requirements only in the across-track and vertical directions.

Since the MWP-SAR system in this article operates in stripmap broadside mode, we take this case as an example and visualize (11), as shown in Fig. 2. As this section analyzes the minimum positioning accuracy requirements in each direction, the squint angle in the cross-track and vertical accuracy requirements is set to 0° for the broadside case, i.e.,

$$\begin{cases} |\Delta x(\eta)| < \dfrac{c}{4B_r\cos\phi} \\ |\Delta z(\eta)| < \dfrac{c}{4B_r\sin\phi}. \end{cases} \tag{12}$$

Fig. 2(a) and (c) shows the cross-track and vertical positioning accuracy requirements as functions of signal bandwidth and depression angle, while Fig. 2(b) shows the along-track requirement as a function of signal bandwidth and squint angle, where the squint angle is determined by the azimuth beamwidth. The white lines in Fig. 2 represent the isograms of these requirements, and the orange line indicates the positioning accuracy of the Applanix POS AV 610 RTX system mounted on the MWP-SAR platform. Table I shows the accuracy of the positioning device [16].

Based on the system parameters in Table II, the red pentagram in Fig. 2 is taken as an example for detailed interpretation. When the signal bandwidth is 5.72 GHz, the

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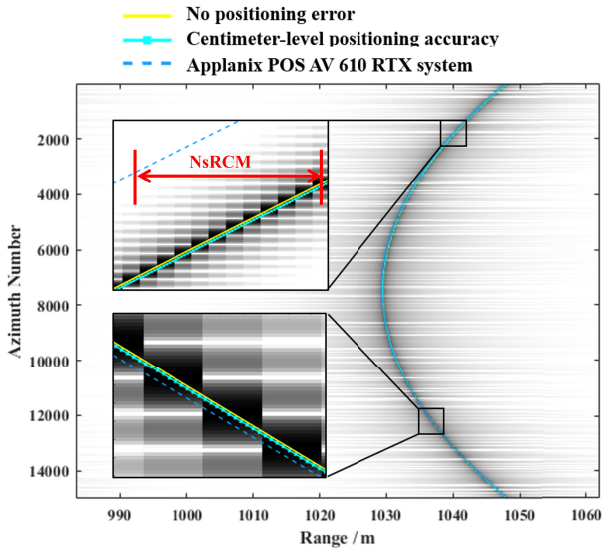


Fig. 3. Range histories obtained from platforms with different positioning accuracies in the pulse-compression domain.

TABLE II
AIRBORNE MWP-SAR SYSTEM PARAMETERS

Parameter	Values
Carrier frequency	15.14 GHz
Signal bandwidth	5.72 GHz
Pulse repetition frequency	10000 Hz
Azimuth beamwidth	30°
Depression angle	25°∼65°
Squint angle	0°
Platform height	500 m
Platform velocity	65.3 m/s

depression angle is 45°, and the target squint angle is 10°, the required positioning accuracy is between 0.01 and 0.02 m in the cross-track and vertical directions, and between 0.05 and 0.1 m in the along-track direction. Fig. 3 further shows the effects of different positioning accuracies in the pulse-compression domain. The colored lines represent the range histories obtained from platforms with varying accuracies. When the platform positioning accuracy is insufficient, NsRCM may even exceed multiple range cells and centimeter-level accuracy is required to keep it within half a range cell.

It is worth noting that satisfying the tolerance of residual motion errors at the range envelope level is insufficient unless the azimuth-phase error is less than $\pi/4$. Thus, the range history error must conform to the following condition, which is more stringent than that in (10):

$$|\Delta R(\eta)| < \frac{c}{16 f_c} = \frac{B_r}{4 f_c}\frac{c}{4 B_r} \tag{13}$$

where $B_r/(4f_c)$ is typically less than 0.5 and equals 0.0945 under the parameters in Table II, indicating that the tolerance at the range envelope level is an order of magnitude larger than at the azimuth phase level.

From the above analysis, it is clear that, for most INS/GPS systems, including the Applanix POS AV 610 RTX, the positioning accuracy exceeds the tolerance of residual motion errors at the range-envelope level, let alone that at the azimuth-phase level. This may even cause NsRCM to exceed multiple range cells, leading to severe 2-D defocusing.

C. Dominant Target Position Error Tolerance Analysis

The trajectory reconstruction method based on dominant targets typically calculates the range history based on the positions of dominant targets in the coarse imaging results and the recorded trajectory. However, as shown in Section II-B, the coarse imaging results are severely 2-D defocused, which makes the dominant target positions highly inaccurate and introduces position errors. The following analyzes the impact of dominant target position errors on trajectory reconstruction and derives their tolerances at the range-envelope and azimuth-phase levels to illustrate the limitation of coarse imaging.

Assume that target P is a dominant target located on the ground. Since dominant target position errors exist, the calculated range history of dominant target P should be expressed as

$$R_{cal}(\eta) = \sqrt{\left(x_i(\eta) - x_p'\right)^2 + \left(y_i(\eta) - y_p'\right)^2 + \left(z_i(\eta) - z_p\right)^2} \tag{14}$$

where (x_p', y_p', z_p) denotes the position of the dominant target obtained from the coarse imaging results. This position is typically determined by identifying the maximum pixel value, and its relation to the real position of the dominant target is given as follows:

$$\left(x_p, y_p, z_p\right) = \left(x_p', y_p', z_p\right) + \left(\delta x_p, \delta y_p, 0\right) \tag{15}$$

where $(\delta x_p, \delta y_p, 0)$ represents the dominant target position error. Since the selected dominant targets are typically ground targets, this article does not consider their position errors in the vertical direction.

By performing a first-order Taylor expansion of the real range history at zero residual motion error and zero dominant target position error, its relation to the calculated range history can be obtained

$$R_r(\eta) = R_{cal}(\eta) + \Delta R'(\eta) + \delta R(\eta) \tag{16}$$

where $\Delta R'(\eta)$ and $\delta R(\eta)$ represent the range history error introduced by residual motion error and dominant target position error, respectively. Their detailed expressions are as follows:

$$\begin{aligned}\Delta R'(\eta) \approx & \left(\frac{x_i(\eta) - x_p'}{R_i'(\eta)}\right)\Delta x(\eta) \\ & + \left(\frac{y_i(\eta) - y_p'}{R_i'(\eta)}\right)\Delta y(\eta) + \left(\frac{z_i(\eta) - z_p}{R_i'(\eta)}\right)\Delta z(\eta)\end{aligned} \tag{17}$$

$$\delta R(\eta) \approx \left(\frac{x_i(\eta) - x_p'}{R_i'(\eta)}\right)\delta x_p + \left(\frac{y_i(\eta) - y_p'}{R_i'(\eta)}\right)\delta y_p. \tag{18}$$

From (9) and (16), it can be seen that when calculating the range history using the recorded ideal trajectory and the dominant target positions obtained from the coarse imaging results, dominant target position errors affect imaging results through both the range envelope and the azimuth phase. This indicates that these errors are coupled with the residual motion

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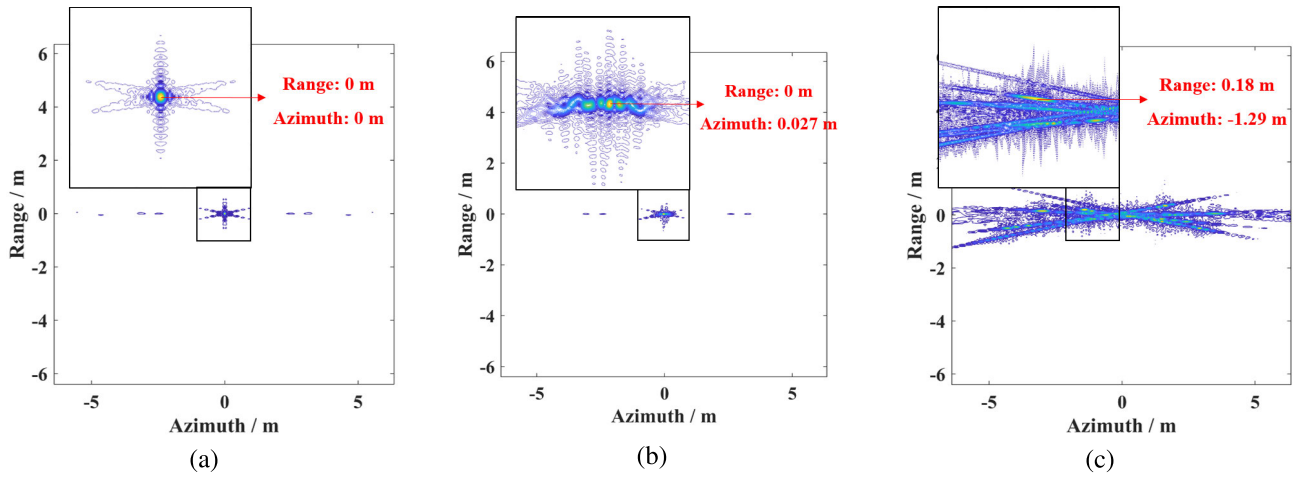


Fig. 4. Imaging results with different positioning accuracies. (a) No positioning error and no NsRCM, (b) centimeter-level positioning accuracy and NsRCM within half a range cell, and (c) Applanix POS AV 610 RTX system and NsRCM greater than half a range cell.

errors, influencing the estimation of residual motion errors and trajectory reconstruction.

Similar to the analysis of the range-envelope-level error tolerance in Section II-B, the absolute value of $\Delta R'(\eta)+\delta R(\eta)$ must be less than $c/(4B_r)$. To analyze the tolerance of dominant target position errors at the range envelope level, $\Delta R'(\eta)$ is assumed to be zero. This assumption is reasonable because as long as $\delta R(\eta)$ is smaller than $c/(4B_r)$, it is still possible to reconstruct a trajectory that keeps the NsRCM within half a range cell. Thus, the range history error induced by dominant target position errors must satisfy the following condition:

$$|\delta R(\eta)| < \frac{c}{4B_r}. \tag{19}$$

By substituting (18) into (19), the dominant target position error tolerance at the range envelope level is expressed as

$$\begin{cases} |\delta x| < \dfrac{c}{4B_r \cos\phi \cos\theta} \\ |\delta y| < \dfrac{c}{4B_r \sin\theta}. \end{cases} \tag{20}$$

Similar to the analysis in (11), the signal bandwidth, the target squint angle, and the depression angle all affect this tolerance, and their influence follows the same trend as that derived from (11).

To highlight the limitations of coarse image, Fig. 4 depicts the coarse images with different positioning accuracies. From these results, it can be observed that when the NsRCM is within half a range cell, the dominant target undergoes only 1-D defocusing, with its position exhibiting minor errors in the azimuth direction. When the NsRCM exceeds half a range cell, the dominant target undergoes severe 2-D defocusing, resulting in significant position errors in both the range and azimuth directions. For most UHR imaging systems, coarse images with 2-D defocusing do not satisfy the dominant target position error tolerance at the range envelope level. As an example, for the MWP-SAR system considered in this article, this tolerance is 0.0145 m in the range direction and 0.0755 m in the azimuth direction, based on the system parameters in Table II. As shown in Fig. 4(c), the range and azimuth errors of 0.18 and −1.29 m, respectively, exceed the corresponding tolerances.

Similar to (13), the dominant target position error tolerance at the azimuth phase level is expressed as

$$|\delta R(\eta)| < \frac{B_r}{4f_c}\frac{c}{4B_r}. \tag{21}$$

This tolerance is typically more than twice that at the range envelope level, as discussed above for (13). When the NsRCM is within half a range cell, the dominant target position exhibits only minor errors, even within a single resolution cell. Therefore, this study accounts for dominant target position errors only at the range envelope level and neglects them at the azimuth phase level, consistent with existing methods [38], [41].

In summary, dominant target position errors affect trajectory reconstruction. The positions obtained from coarse images with 2-D defocusing do not satisfy the range-envelope-level error tolerance, while the corresponding errors at the azimuth phase level can be neglected.

III. Full-Trajectory Reconstruction

The proposed full-trajectory reconstruction method is detailed in this section, and its overall flowchart is shown in Fig. 5. Section III-A introduces the preprocessing stage, including coarse imaging and dominant target extraction. Sections III-B and III-C describe the stepwise trajectory reconstruction, consisting of range-envelope-level trajectory reconstruction followed by azimuth-phase-level trajectory reconstruction. This part explains in detail how the high-dimensional optimization problem, which jointly involves residual motion errors and dominant target position errors, is reduced to a low-dimensional one by optimizing only the latter. Finally, Section III-D describes the procedure for optimally fusing multiple overlapping subtrajectories based on the minimum waveform entropy criterion.

A. Preprocess

The objective of preprocessing is to extract multiple dominant targets for subsequent trajectory reconstruction. This

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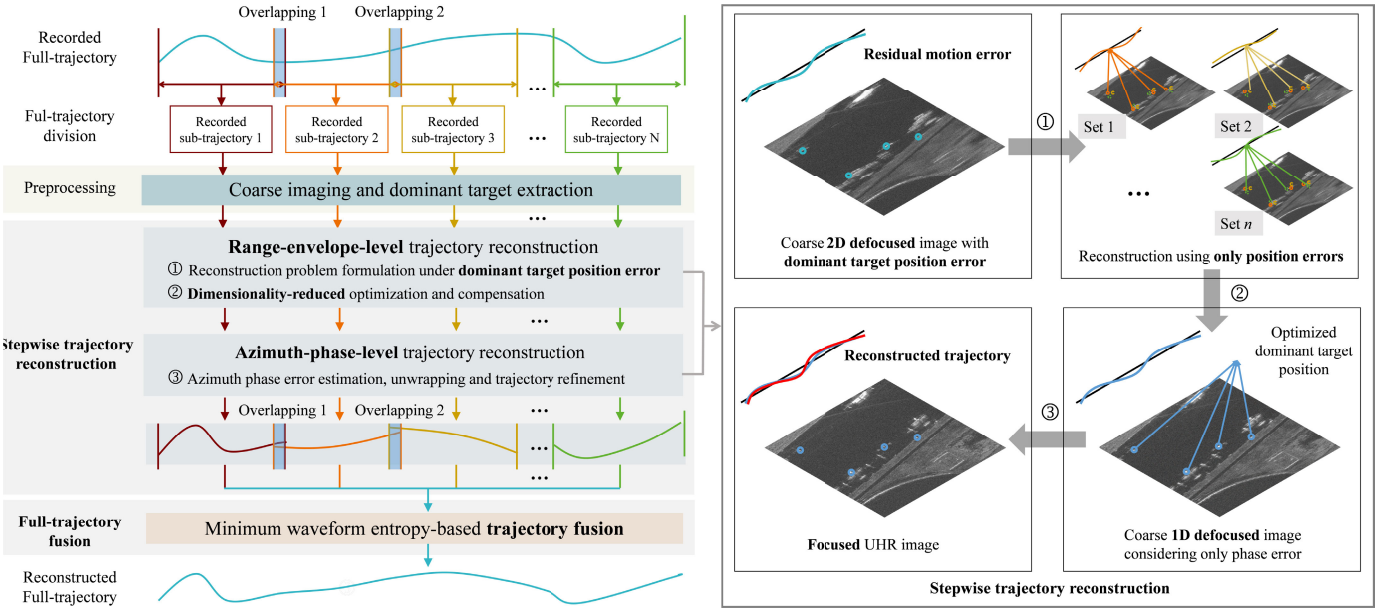


Fig. 5. Flowchart of the proposed full-trajectory reconstruction method.

section consists of two parts: coarse imaging and dominant target extraction.

1) Coarse Imaging: Both TDAs and FDAs can be used for coarse imaging. In general, FDAs are more computationally efficient than TDAs, but TDAs are adopted in this article. The process of BPA is as follows:

$$s_{BP} = \sum_{\eta} s_{r_deramp}(r_\tau, \eta) \exp\left(j\frac{4\pi f_c}{c} R_{cal}(\eta)\right). \tag{22}$$

The choice of TDAs is motivated by three considerations: 1) FDAs must be combined with motion error compensation algorithms to achieve imaging quality comparable to TDAs, which increases computational burden; 2) dominant target position extracted from frequency-domain images requires conversion to ground-range–azimuth coordinates, while those from time-domain images are already in that form; and 3) coarse imaging imposes relatively low grid-size requirements for TDAs, resulting in fewer grid points and hence lower computational cost. Moreover, the implementation of BPA on GPUs [43], together with the development of accelerated algorithms such as fast factorized backprojection (FFBP) [44], further motivates the choice of TDAs in this work.

2) Dominant Target Extraction: The coarse imaging result serves as the basis for dominant target extraction. This study defines a dominant target as an extractable single-point target, either an isolated point or a strong point within a weak target region. Such a target appears in the pulse-compression domain as an extractable RCM. Therefore, high signal-to-clutter-ratio (SCR) regions are expected to be selected for dominant target extraction. Moreover, the method proposed in [45] can be used for efficient and automated extraction of dominant targets. It is preferable to select dominant targets on the ground plane to improve the trajectory reconstruction accuracy.

B. Range-Envelope-Level Trajectory Reconstruction

The purpose of this reconstruction step is to ensure that the NsRCM does not exceed half a range cell, reducing image defocusing from 2-D to 1-D and thereby facilitating subsequent phase error estimation and further trajectory refinement.

1) Range History Error Minimization Problem Formulation: To achieve this goal, the real RCM of the dominant targets must be extracted as prior information. This process can be performed following the energy-trajectory extraction method proposed by Hai et al. [46]. Then, for any given dominant target, the following equation can be established:

$$R_r^{\mathrm{RE}}(\eta) = R_{\mathrm{cal}}^{\mathrm{RE}}(\eta) + \Delta R'^{\mathrm{RE}}(\eta) + \delta R^{\mathrm{RE}}(\eta) \tag{23}$$

where the superscript RE stands for range envelope; $R_r^{\mathrm{RE}}(\eta)$ denotes the real RCM or range-envelope-level range history of the dominant target; $R_{\mathrm{cal}}^{\mathrm{RE}}(\eta)$ represents the range history calculated from the recorded trajectory and the dominant target positions obtained from the coarse image; and $\Delta R'^{\mathrm{RE}}(\eta)$ and $\delta R^{\mathrm{RE}}(\eta)$ denote the range-envelope-level range history errors induced by the residual motion errors and the dominant target position errors, respectively. Their expressions are similar to that of (17) and (18).

It should be emphasized that the RCM obtained from energy-trajectory extraction is discrete, and $R_r^{\mathrm{RE}}(\eta)$ in the above equation corresponds to the RCM after classical Gaussian smoothing.

Based on (23), the range-envelope-level trajectory reconstruction is the process of solving this equation. Clearly, more dominant targets are required to obtain additional equations for trajectory reconstruction. Suppose the number of azimuth points and dominant targets is N and K, respectively. Then, the number of unknowns is $3N$ (residual motion errors) $+2K$ (dominant target position errors), while the number of equations is KN. Therefore, the following condition must be

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satisfied to ensure that the equations are solvable:

$$KN \geq 3N + 2K. \tag{24}$$

Consequently, the number of dominant targets K must satisfy the following relation:

$$K \geq \left\lceil \frac{3N}{N-2} \right\rceil > 3. \tag{25}$$

When $K \geq 4$, the trajectory reconstruction using the real RCMs of K dominant targets can be formulated as the following solvable range history error minimization problem:

$$\min_{\mathbf{R},\mathbf{\Lambda}} \sum_{d=1}^{K} \sqrt{\sum_{\eta=1}^{N} \left(\Delta R_r^{\mathrm{RE}}(\eta; P_d) - \widehat{R}^{\mathrm{RE}}(\eta; P_d; \mathbf{R}, \mathbf{\Lambda}) \right)^2} \tag{26}$$

where $\mathbf{R}$ denotes the residual motion error vector over all azimuth time instants, $\mathbf{\Lambda}$ represents the dominant target position error vector, and

$$\Delta R_r^{\mathrm{RE}}(\eta; P_d) = R_r^{\mathrm{RE}}(\eta; P_d) - R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_d) \tag{27}$$

$$\widehat{R}^{\mathrm{RE}}(\eta; P_d; \mathbf{R}, \mathbf{\Lambda}) = \Delta R'^{\mathrm{RE}}(\eta; P_d; \mathbf{R}) + \delta R^{\mathrm{RE}}(\eta; P_d; \mathbf{\Lambda}). \tag{28}$$

The optimization problem in (26) is high-dimensional, and directly solving it is computationally complex and time-consuming. The following derivation details how this problem is reduced to a lower dimensional form in this work, enabling more efficient trajectory reconstruction.

2) Dimensionality-Reduced Solution: For a given azimuth time, a system of equations composed of K equations from (23) can be obtained. By substituting (17) and (18) into $\Delta R'^{\mathrm{RE}}(\eta; P_d; \mathbf{R})$ and $\delta R^{\mathrm{RE}}(\eta; P_d; \mathbf{\Lambda})$, the complete form of the equations is derived. For convenience in subsequent derivation, the system is expressed in the following form:

$$\mathbf{Y} = \mathbf{\Omega E} + \mathbf{H\Lambda} \tag{29}$$

where $\mathbf{Y}$ represents the difference between the real RCM of the dominant target and the calculated range-envelope-level range history, expressed as follows:

$$\mathbf{Y} = \left[\Delta R_r^{\mathrm{RE}}(\eta; P_1), \Delta R_r^{\mathrm{RE}}(\eta; P_2), \ldots, \Delta R_r^{\mathrm{RE}}(\eta; P_K) \right]_{K\times 1}^{\top}. \tag{30}$$

In addition, $\mathbf{H}$ denotes the measurement matrices corresponding to the residual motion errors, expressed as

$$\mathbf{H} = [\mathbf{H}_1, \mathbf{H}_2]_{K\times 2K} \tag{31}$$

where

$$\begin{cases} \mathbf{H}_1 = \mathrm{diag}\left\{ \frac{x_i(\eta) - x_1'}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_1)}, \ldots, \frac{x_i(\eta) - x_K'}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_K)} \right\} \\ \mathbf{H}_2 = \mathrm{diag}\left\{ \frac{y_i(\eta) - y_1'}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_1)}, \ldots, \frac{y_i(\eta) - y_K'}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_K)} \right\}. \end{cases} \tag{32}$$

$\mathbf{\Omega}$ denotes the measurement matrices corresponding to the dominant target position errors, expressed as

$$\mathbf{\Omega} = \begin{bmatrix} \frac{x_i(\eta) - x_1'}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_1)} & \frac{y_i(\eta) - y_1'}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_1)} & \frac{z_i(\eta) - z_1}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_1)} \\ \frac{x_i(\eta) - x_2'}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_2)} & \frac{y_i(\eta) - y_2'}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_2)} & \frac{z_i(\eta) - z_2}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_2)} \\ \vdots & \vdots & \vdots \\ \frac{x_i(\eta) - x_K'}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_K)} & \frac{y_i(\eta) - y_K'}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_K)} & \frac{z_i(\eta) - z_K}{R_{\mathrm{cal}}^{\mathrm{RE}}(\eta; P_K)} \end{bmatrix}_{K\times 3}. \tag{33}$$

The remaining parameters, $\mathbf{E}$ and $\mathbf{\Lambda}$, denote the residual motion errors and dominant target position errors, respectively, with expressions given as follows:

$$\mathbf{E} = \left[\Delta x(\eta), \Delta y(\eta), \Delta z(\eta) \right]^{\top} \tag{34}$$

$$\mathbf{\Lambda} = \left[\mathbf{\Lambda}_1; \mathbf{\Lambda}_2 \right]_{2K\times 1} \tag{35}$$

where

$$\begin{cases} \mathbf{\Lambda}_1 = \left[\delta x_1, \delta x_2, \ldots, \delta x_K \right]^{\top} \\ \mathbf{\Lambda}_2 = \left[\delta y_1, \delta y_2, \ldots, \delta y_K \right]^{\top}. \end{cases} \tag{36}$$

The error vectors $\mathbf{E}$ and $\mathbf{\Lambda}$ are coupled, and they jointly induce the range history error. Through an LS approach, the problem can be simplified, after which one can be expressed in terms of the other. Specifically, the residual motion error at any azimuth time can be represented by the dominant target position error, as given by

$$\mathbf{E} = \left(\mathbf{\Omega}^{\top} \mathbf{\Omega} \right)^{-1} \mathbf{\Omega}^{\top} (\boldsymbol{Y} - \boldsymbol{H\Lambda}). \tag{37}$$

Since $\boldsymbol{R}$ can be represented in terms of $\mathbf{\Lambda}$, the reconstructed trajectory and its corresponding range history can also be expressed using $\mathbf{\Lambda}$. Therefore, (26) can be reformulated as the following optimization problem involving only $\mathbf{\Lambda}$:

$$\min_{\mathbf{\Lambda}} \left[\sum_{d=1}^{K} \sqrt{\sum_{\eta=1}^{N} \left(R_r^{\mathrm{RE}}(\eta; P_d) - \left\| \mathbf{E} + \mathbf{R} - \mathbf{T}_{P_d} \right\|_2 \right)^2} \right] + \lambda_1 \|\mathbf{\Lambda}_1\|_2 + \lambda_2 \|\mathbf{\Lambda}_2\|_2 \tag{38}$$

where $\mathbf{R} = (x_i(\eta), y_i(\eta), z_i(\eta))^{\top}$ denotes the recorded trajectory, $\mathbf{T}_{P_d} = (x_{P_d}', y_{P_d}', z_{P_d})^{\top} + (\mathbf{\Lambda_1}, \mathbf{\Lambda}_2, 0)^{\top}$ denotes the estimated dominant target position, and λ_1 and λ_2 are regularization coefficients.

To reduce manual parameter tuning and keep the regularization strength consistent across scenes, λ_1 and λ_2 are determined in a scale-consistent manner based on the coarse imaging initialization $\mathbf{\Lambda}^{(0)}$. Let $J_1(\mathbf{\Lambda})$ denote the data term in (38). The regularization coefficients are specified as

$$\lambda_1 = \rho_1 \frac{J_1\left(\mathbf{\Lambda}^{(0)}\right)}{\left\| \mathbf{\Lambda}_1^{(0)} \right\|_2 + \varepsilon} \qquad \lambda_2 = \rho_2 \frac{J_1\left(\mathbf{\Lambda}^{(0)}\right)}{\left\| \mathbf{\Lambda}_2^{(0)} \right\|_2 + \varepsilon} \tag{39}$$

where ρ_1 and ρ_2 are dimensionless weighting factors controlling the overall regularization level and ε is a small constant introduced to avoid numerical singularities. Furthermore, the ratios λ_1 and λ_2 are selected according to the relative defocus severity in azimuth and range inferred from the coarse imaging result. For example, when the degree of defocusing in azimuth surpasses that in range, the condition $\lambda_1 > \lambda_2$ holds.

Through the above procedure, the dimensionality of the optimization parameters is reduced from $3N + 2K$ to $2K$. This reduction simplifies the optimization process, enabling optimal reconstruction of the range-envelope-level trajectory solely by adjusting the positions of the dominant targets.

The objective in (38) is differentiable but generally nonconvex. The adaptive moment estimation (Adam) algorithm is a widely used first-order method. In practice, it has demonstrated strong performance on differentiable nonconvex objectives by leveraging adaptive step sizes and momentum, so it is adopted

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Algorithm 1 Adam-Based Target Position Error Optimization

Initialize: $\mathbf{\Lambda}$ (target position error), step size α, exponential decay rates $\beta_1, \beta_2 \in [0, 1)$, small constant $\epsilon = 10^{-8}$
Initialize: first moment vector $m_0 = 0$, second moment vector $v_0 = 0$
for each iteration $i = 1, 2, \ldots, itermax$ **do**
  Compute gradient $g_i = \nabla J(\mathbf{\Lambda}_{i-1})$
  Update biased first moment estimate: $m_i = \beta_1 m_{i-1} + (1-\beta_1) g_i$
  Update biased second moment estimate: $v_i = \beta_2 v_{i-1} + (1-\beta_2) g_i^2$
  Compute bias-corrected first moment: $\hat{m}_i = \frac{m_i}{1-\beta_1^i}$
  Compute bias-corrected second moment: $\hat{v}_i = \frac{v_i}{1-\beta_2^i}$
  Update parameters: $\mathbf{\Lambda}_i = \mathbf{\Lambda}_{i-1} - \alpha \frac{\hat{m}_i}{\sqrt{\hat{v}_i}+\epsilon}$
end for

to enhance efficiency, convergence, and robustness [47]. The derivation is outlined below and the whole optimization process is shown in Algorithm 1.

Let $J(\mathbf{\Lambda})$ represent (38). Through derivation, the first derivative of $J(\mathbf{\Lambda})$ with respect to $\mathbf{\Lambda}$ can be obtained, and the expression is as follows:

$$\nabla J(\mathbf{\Lambda}) = \sum_{d=1}^{K}\sum_{\eta=1}^{N} \frac{\left(1 - R_r^{\mathrm{RE}}(\eta; P_d)/\|\mathbf{\Gamma}\|_2\right)}{\sqrt{\sum_\eta \left(R_r^{\mathrm{RE}}(\eta; P_d) - \|\mathbf{\Gamma}\|_2\right)^2}} \mathbf{\Gamma}^\top (\mathbf{Q} - \mathbf{\Theta}) + \left(\frac{\lambda_1 \mathbf{\Lambda_1}}{\|\mathbf{\Lambda_1}\|_2}; \overrightarrow{\mathbf{0}}\right) + \left(\overrightarrow{\mathbf{0}}; \frac{\lambda_2 \mathbf{\Lambda}_2}{\|\mathbf{\Lambda}_2\|_2}\right) \tag{40}$$

where $\overrightarrow{\mathbf{0}}$ is a column vector of size $K \times 1$, with all entries being 0. $\mathbf{\Gamma}$ and $\mathbf{Q}$ are expressed as follows:

$$\begin{cases} \mathbf{\Gamma} = \mathbf{E} + \mathbf{M} - \mathbf{T}_{P_d} \\ \mathbf{Q} = \left(\mathbf{\Omega}^\top \mathbf{\Omega}\right)^{-1} \mathbf{\Omega}^\top \mathbf{H}. \end{cases} \tag{41}$$

$\mathbf{\Theta}$ is a $3 \times 2K$ matrix in which only the first and second rows of the dth column are equal to one, while all other entries are zero, expressed as

$$\mathbf{\Theta} = \begin{bmatrix} 0 & \cdots & 1 & \cdots & 0 \\ 0 & \cdots & 1 & \cdots & 0 \\ 0 & \cdots & 0 & \cdots & 0 \end{bmatrix}_{3\times 2K}. \tag{42}$$

Additionally, a multistart strategy can be adopted when a single initialization fails to yield a satisfactory solution. Algorithm 1 is first run using one coarse-image-based initialization. If the optimization terminates with an unsatisfactory outcome, Algorithm 1 is re-executed from multiple distinct coarse-image-based initializations, and the solution attaining the minimum objective value is selected. This workflow is consistent with standard multistart frameworks for enhancing robustness in nonconvex optimization problems [48].

By substituting the optimally estimated $\hat{\mathbf{\Lambda}}$ into (37), the residual motion error vector $\hat{\mathbf{E}}$ at all azimuth time instants is obtained. Compensating these errors to the recorded trajectory yields the reconstructed range-envelope-level trajectory.

C. Azimuth-Phase-Level Trajectory Reconstruction

The objective of this reconstruction step is to bridge the accuracy gap between the range envelope level and the azimuth phase level, thereby refining the trajectory. This section is divided into two parts: azimuth-phase error estimation and trajectory refinement.

1) Azimuth-Phase Error Estimation: The trajectory reconstructed in Section III-B reduces the image defocusing from 2-D to 1-D. Under this 1-D defocusing, the azimuth-phase error of the dominant targets can be estimated via autofocus methods [35], [38] or computed from the real azimuth phase extracted in the pulse-compression domain. This estimation process is significantly simplified owing to the stepwise reconstruction strategy.

However, to ensure continuity for subsequent trajectory refinement, the estimated azimuth-phase errors need to be unwrapped. According to (3), the azimuth-phase difference between adjacent azimuth times is expressed as

$$\Delta\psi_{diff} = \frac{4\pi \Delta R_{\mathrm{diff}}}{\lambda} \approx \frac{4\pi V_r \sin\theta}{\lambda f_{\mathrm{PRF}}} < 2\pi. \tag{43}$$

When designing an SAR system, the PRF should be set higher than the Doppler frequency. Substituting this condition into (43) yields that (43) is less than (2π). Specifically, based on the system parameters listed in Table II, the maximum azimuth-phase difference between adjacent azimuth times is 0.7191, which does not exceed π. It does not cause phase ambiguity, so phase unwrapping can be easily performed. For other systems with more stringent parameters, phase unwrapping can be performed by referring to the methods developed in [49] and [50].

2) Trajectory Refinement: In the azimuth-phase-level trajectory reconstruction, the effect of the dominant target position errors can be neglected, as analyzed in Section II-C and done in existing methods [38], [41]. Let $\Delta\psi(\eta; P_d)$ denote the estimated azimuth-phase error of dominant target P_d, whose corresponding range history error is $\Delta R^{\mathrm{AP}}(\eta; P_d)$. The estimation of the residual motion error $\mathbf{E}_{\mathrm{AP}}$ in the azimuth-phase-level reconstruction then is expressed as

$$\hat{\mathbf{E}}_{\mathrm{AP}} = \arg\min_{\mathbf{E}_{\mathrm{AP}}} \|\mathbf{Y}_{\mathrm{AP}} - \mathbf{\Omega}_{\mathrm{AP}} \mathbf{E}_{\mathrm{AP}}\|_2 \tag{44}$$

where $\mathbf{Y}_{\mathrm{AP}}$ is a column vector composed of $\Delta R^{\mathrm{AP}}(\eta; Pd)$ and $\mathbf{\Omega}_{\mathrm{AP}}$ is obtained by replacing the recorded trajectory in (33) with the range-envelope-level reconstructed trajectory, with dominant target positions updated to their optimized values. $\mathbf{E}_{\mathrm{AP}}$ denotes the azimuth-phase-level residual motion error at a specific azimuth time instant, expressed as

$$\mathbf{E}_{\mathrm{AP}} = \left[\Delta x_{\mathrm{AP}}(\eta), \Delta y_{\mathrm{AP}}(\eta), \Delta z_{\mathrm{AP}}(\eta)\right]^\top. \tag{45}$$

Using the LS method to solve (44), yielding the following expression:

$$\hat{\mathbf{E}}_{\mathrm{AP}} = \left(\mathbf{\Omega}_{\mathrm{AP}}{}^\top \mathbf{\Omega}_{\mathrm{AP}}\right)^{-1} \mathbf{\Omega}_{\mathrm{AP}}{}^\top \mathbf{Y}_{\mathrm{AP}}. \tag{46}$$

By compensating the residual motion error obtained from (46) into the range-envelope-level reconstructed trajectory, the azimuth-phase-level trajectory reconstruction is achieved. Additionally, for low-resolution cases where the NsRCM does

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not exceed half a range cell, azimuth-phase-level trajectory reconstruction can be performed directly, without resorting to range-envelope-level trajectory reconstruction.

D. Full-Trajectory Fusion

Although the trajectory has been refined at the azimuth phase level, the reconstructed length is limited to a single full-aperture. In the spotlight imaging mode, the reconstructed trajectory can cover the entire scene. However, in the stripmap imaging mode, it is necessary to fuse multiple subtrajectories to achieve full scene coverage.

Zero- and first-order deviations exist among the independently reconstructed subtrajectories, as such deviations do not affect the focusing quality. The adopted deviation correction strategy is to introduce overlaps between adjacent subtrajectories and exploit the information within the overlapping segments to correct them. The overlap ratio is typically set to about 20%, balancing fusion accuracy and computational efficiency. The specific procedure is as follows.

Let the two adjacent reconstructed subtrajectories be denoted as $\mathbf{K_1}(\boldsymbol{\eta_1})$ and $\mathbf{K_2}(\boldsymbol{\eta_2})$, of sizes $3 \times N_1$ and $3 \times N_2$, respectively. By aligning one subtrajectory with the other through rotation and translation of its overlapping portion, the transformed subtrajectory is obtained as

$$\mathbf{K_{1_rot}}\left(\boldsymbol{\eta_1}\right) = \mathbf{R_x}\mathbf{R_z}\mathbf{K_1}\left(\boldsymbol{\eta_1}\right) + \mathbf{C} \tag{47}$$

where the rotation matrices $\mathbf{R_x}$ and $\mathbf{R_z}$ are given by

$$\mathbf{R_x} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix} \tag{48}$$

$$\mathbf{R_z} = \begin{bmatrix} \cos\beta & -\sin\beta & 0 \\ \sin\beta & \cos\beta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{49}$$

and the translation vector $\mathbf{C}$ is denoted as

$$\mathbf{C} = \begin{bmatrix} c_x & c_y & c_z \end{bmatrix}^{\top}. \tag{50}$$

The overlapping segments $\mathbf{K_1}(\boldsymbol{\eta}_{\mathrm{ol}})$ and $\mathbf{K_2}(\boldsymbol{\eta}_{\mathrm{ol}})$, of size $3 \times N_{\mathrm{ol}}$, are fused using a weighted approach as follows:

$$\mathbf{K_{fusion}}\left(\boldsymbol{\eta_{ol}}\right) = \mathbf{K_{1_rot}}\left(\boldsymbol{\eta_{ol}}\right)\mathbf{W_1} + \mathbf{K_2}\left(\boldsymbol{\eta_{ol}}\right)\mathbf{W_2} \tag{51}$$

where $\mathbf{W_1}$ and $\mathbf{W_2}$ are weighting matrices, with their explicit forms given by

$$\begin{cases} \mathbf{W}_1 = \mathrm{diag}\left\{w_1, w_2, \ldots, w_{N_{ol}}\right\} \\ \mathbf{W}_2 = \mathrm{diag}\left\{1 - w_1, 1 - w_2, \ldots, 1 - w_{N_{ol}}\right\}. \end{cases} \tag{52}$$

Linear weights are employed for subtrajectory fusion. To maintain continuity in the fused trajectory, the weights are chosen as $w_1 = 1$ and $w_{N_{ol}} = 0$.

To optimally fuse the subtrajectories, phases of the dominant targets whose apertures cover the overlapping segments are extracted. The rotation and translation parameters are then estimated by maximizing the similarity between the extracted phases and the phases calculated from the fused trajectory. Inspired by autofocusing methods that employ IE as a metric, this study adopts the waveform entropy of the cross correlation between the two phases as the similarity metric, which better conforms to SAR imaging characteristics. For selected dominant target P_d, the phase cross correlation is

$$y_{cor}\left(k; P_d\right) = \sum_{\eta} \frac{4\pi R_{fusion}\left(\eta + k; P_d\right)\varphi\left(\eta; P_d\right)}{\lambda} \tag{53}$$

where $R_{\mathrm{fusion}}(\eta + k; P_d)$ is the range history derived from the fused trajectory and dominant target P_d and $\varphi(\eta; P_d)$ is the extracted azimuth phase of P_d.

Minimizing the entropy of the waveform in (52) yields the optimal estimation of the rotation and translation parameters. Typically, multiple dominant targets are jointly used to obtain more accurate estimates, as expressed in the following:

$$\min_{\alpha,\beta,\mathbf{C}} -\sum_{d}\sum_{k} \frac{y_{cor}\left(k; P_d\right)^2}{\sum_k y_{cor}\left(k; P_d\right)^2} \log\left(\frac{y_{cor}\left(k; P_d\right)^2}{\sum_k y_{cor}\left(k; P_d\right)^2}\right). \tag{54}$$

The above optimization problem is nonlinear and involves logarithmic functions, making differentiation complex. Given the small number of parameters and the availability of reliable initial parameter values, this article uses particle swarm optimization (PSO) to solve the optimization problem.

Trajectory fusion offers an additional benefit. In general, the full trajectory is segmented into multiple subtrajectories according to the synthetic aperture length. However, if a dominant target's aperture is smaller than the full-aperture-length, the subtrajectory can be subdivided, with each segment covering a subaperture. This strategy improves the general applicability of the proposed reconstruction method and alleviates the conditions imposed on dominant targets.

IV. Experimental Results

To validate the effectiveness of the proposed full-trajectory reconstruction method, both simulation and measured data experiments are conducted. Section IV-A presents the simulation results in terms of reconstruction accuracy, robustness to initialization, sensitivity to the regularization parameters, and point target imaging results. Section IV-B presents the imaging results of measured data.

A. Results on Simulated Data

Based on the system parameters listed in Table II, which are consistent with those of the real MWP-SAR system in Section IV-B, 400 point targets are uniformly distributed within a 400×400 m area. The imposed 3-D residual motion errors, that is, the difference between the real and recorded ideal trajectory, are shown in Fig. 6(b).

1) Trajectory Reconstruction Accuracy: The real trajectory, recorded ideal trajectory, and reconstructed trajectory are simultaneously presented in Fig. 6(a). Fig. 6(c) further shows their differences in the across-track, along-track, and vertical directions. These results demonstrate that the reconstructed trajectory achieves submillimeter-level accuracy, which satisfies the focusing requirements of most systems, as discussed in Section II-B on the tolerance to residual motion errors.

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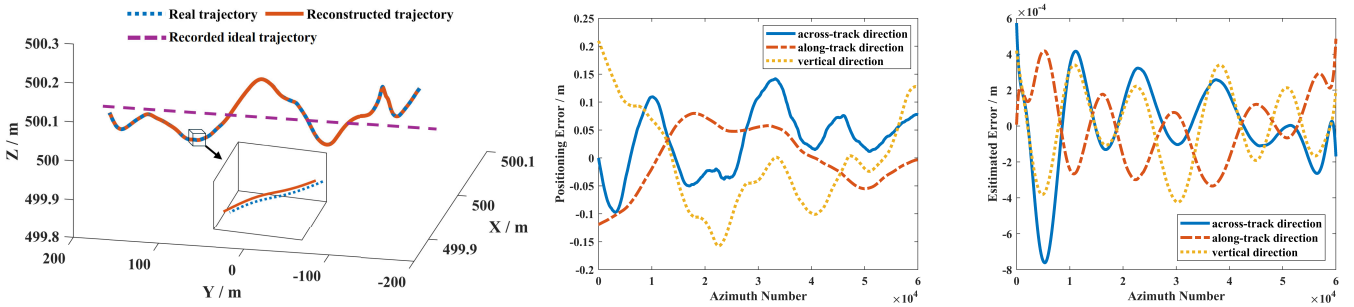


Fig. 6. Recorded ideal trajectory, real trajectory, and reconstructed trajectory, along with their details. (a) Three trajectories and their magnified views. (b) Residual motion errors. (c) Difference between the reconstructed and real trajectories.

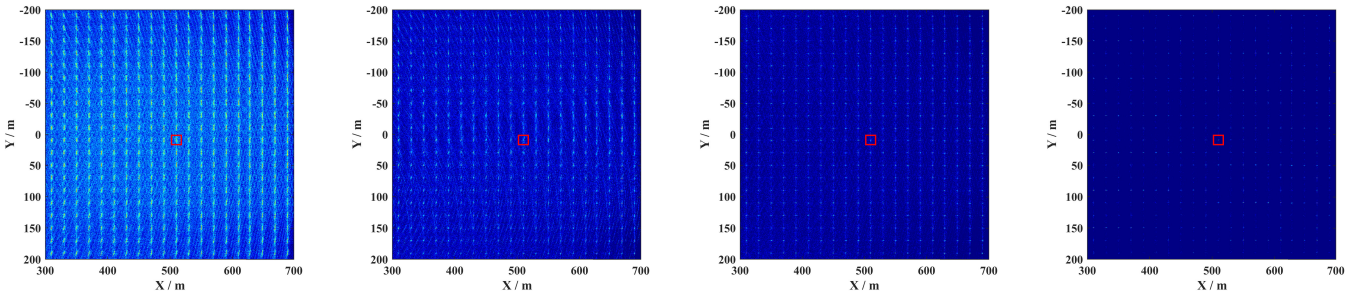


Fig. 7. Imaging results of different processing schemes. (a) Without autofocus, (b) with conventional BP autofocus [36], (c) MWP-SAR imaging method in [16], and (d) proposed method.

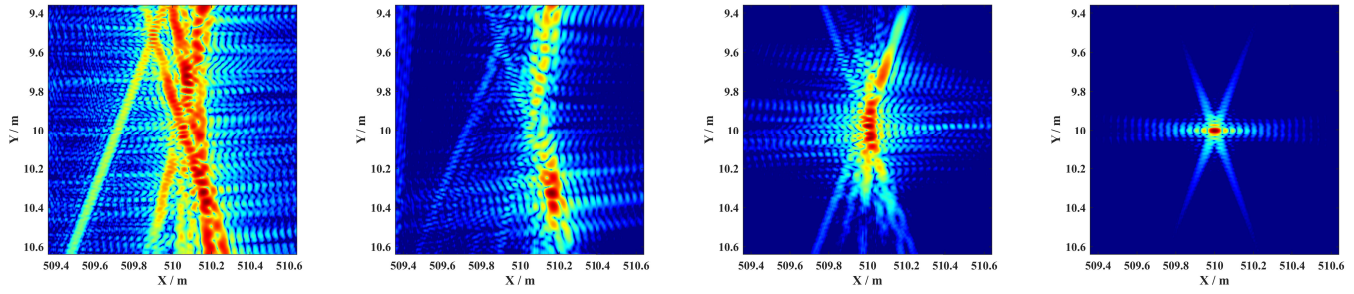


Fig. 8. Imaging results at the scene center of different processing schemes. (a) Without autofocus, (b) with conventional BP autofocus [36], (c) MWP-SAR imaging method in [16], and (d) proposed method.

2) Simulated Imaging Results: To evaluate the effectiveness of the proposed method for UHR large-scene imaging, the simulation progressively compares four processing schemes: without autofocus, with conventional BP autofocus [36], the MWP-SAR imaging method in [16], and the proposed method. Specifically, the first scheme performs imaging using only the recorded trajectory, corresponding to the case without autofocus, to examine whether the recorded trajectory is sufficiently accurate for UHR imaging. The second scheme applies conventional BP autofocus to the image formed from the recorded trajectory, investigating whether compensating only spatial invariant phase errors is sufficient under this condition. The third scheme adopts the MWP-SAR imaging method in [16], which reconstructs the trajectory through optimal subaperture division to account for spatial-variant residual motion errors, aiming to verify whether this existing MWP-SAR imaging method is adequate for the UHR scenario considered in this article. The BPA is utilized as the common imaging algorithm for all schemes.

Fig. 7 shows the imaging results obtained by the different schemes, and Fig. 8 provides the enlarged views of the red boxed region in Fig. 7. As shown in Fig. 7(a), imaging without autofocus produces severe defocusing in both range and azimuth. This indicates that, under the UHR system configuration considered here, the recorded trajectory is not accurate enough to keep the NsRCM within half a range cell. Residual motion error compensation is therefore indispensable. The result of imaging with conventional BP autofocus is shown in Fig. 7(b). Although the focusing quality is improved relative to Fig. 7(a), noticeable defocusing remains. This indicates

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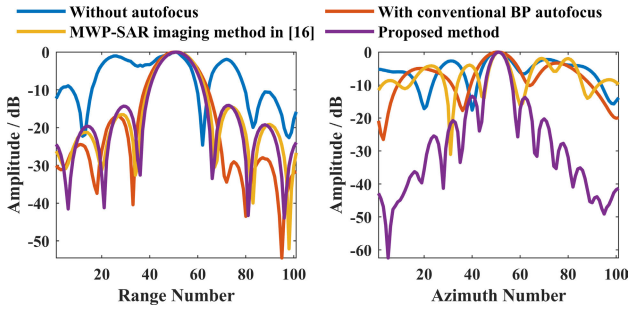


Fig. 9. Range and azimuth profiles at the scene center obtained using different processing schemes.

TABLE III

PERFORMANCE COMPARISON OF DIFFERENT PROCESSING SCHEMES AT THE SCENE CENTER IN TERMS OF RANGE AND AZIMUTH METRICS

Scheme	Range profile			Azimuth profile		
	IRW (m)	PSLR (dB)	ISLR (dB)	IRW (m)	PSLR (dB)	ISLR (dB)
Without autofocus	0.058	-2.500	-3.819	0.075	-5.367	-4.921
Conventional BP autofocus	0.038	-22.19	-16.99	0.034	-8.505	0.463
MWP-SAR method in [16]	0.036	-15.87	-12.59	0.017	-9.512	4.654
Proposed method	0.033	-14.84	-11.93	0.017	-14.14	-11.80

that, in the considered UHR large-scene imaging case, spatial-variant residual motion errors cannot be adequately handled by the conventional BP autofocus procedure that compensates only spatial invariant phase errors. Fig. 7(c) shows the result obtained with the method in [16]. By reconstructing the trajectory, the spatial-variant nature of the residual motion error is taken into account, and the focusing performance is further improved. Nevertheless, residual defocusing can still be observed. This is because the subimage registration procedure employed in [16] becomes more challenging in large-scene imaging, which limits the achievable imaging performance. The result obtained with the proposed method is shown in Fig. 7(d). Using the reconstructed trajectory, the

image is well focused. This demonstrates that the proposed method can effectively resolve the imaging degradation caused by spatial-variant residual motion errors in UHR large-scene imaging.

To more reliably validate the effectiveness of the proposed method, this article presents the range and azimuth profiles at the scene center obtained by different processing schemes, shown in Fig. 9, and quantitatively derives the corresponding impulse response width (IRW), peak sidelobe ratio (PSLR), and integral sidelobe ratio (ISLR) metrics, which are summarized in Table III. Note that all ranges mentioned in the simulations and measured data results refer to ground range. Among the evaluated processing schemes, the proposed method provides desirable IRWs in both range and azimuth while maintaining reasonable PSLR and ISLR values. These quantitative results are consistent with the visual focusing performance and further verify that the reconstructed trajectory satisfies the focusing requirement in the considered UHR large-scene simulation.

3) Robustness to Initialization: To evaluate the effect of the initialization of the dominant target position error on trajectory reconstruction accuracy, 100 Monte Carlo trials are conducted. In each trial, the initial dominant target position

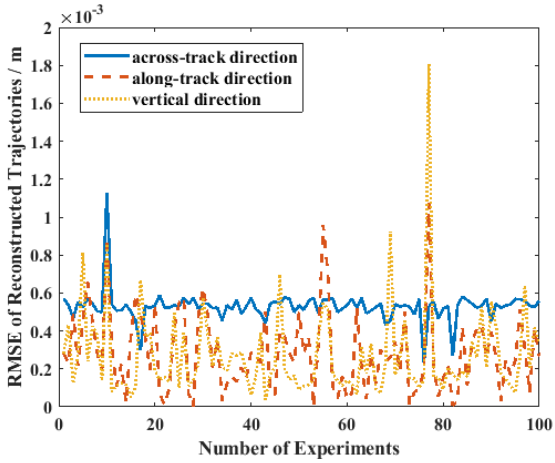


Fig. 10. RMSE results of the reconstructed trajectories obtained from 100 different initializations.

error is generated by random sampling within the 3-dB energy contour of the corresponding dominant target in the coarse image. This initialization rule is physically reasonable because, under defocus and noise, the target energy in the coarse image is mainly concentrated within the 3-dB region, which provides a practical bound on the uncertainty of the peak location. As a result, sampling within this region yields feasible initial values that are consistent with the accuracy typically available from coarse imaging.

Fig. 10 reports the results of 100 Monte Carlo trials by plotting the trajectory reconstruction root-mean-square error (RMSE) versus the trial index in the along-track, cross-track, and vertical directions. With the initialization restricted to the 3-dB energy contour in the coarse image and all other experimental settings unchanged, the RMSEs exhibit limited variation across trials and remain at the millimeter level. This result suggests a low sensitivity to initialization within the considered, practically reasonable range.

4) Sensitivity to Regularization Parameters: This section evaluates the sensitivity of the proposed method to the dimensionless weights ρ_1 and ρ_2 used in (39). Since λ_1 and λ_2 are determined from the initialization $\mathbf{\Lambda}^{(0)}$ via (39), the resulting regularization strength is normalized by both the data term magnitude $J_1(\mathbf{\Lambda}^{(0)})$ and the initial offset norms, which is intended to reduce scene-dependent manual tuning.

A grid-search simulation is performed by independently sweeping ρ_1 and ρ_2 from 0 to 1 with an increment of 0.1, yielding 121 parameter combinations. For each (ρ_1, ρ_2) pair, the same reconstruction pipeline is executed with all other settings kept unchanged. The performance is quantified by the RMSE of the reconstructed trajectory in the along-track, cross-track, and vertical directions, as shown in Fig. 11.

The results demonstrate that the trajectory RMSE is more sensitive to ρ_1 than to ρ_2. This is expected because the range-direction defocus in this simulation is relatively small. In this case, the data term provides a weaker correction drive in the range-related component. If ρ_1 is too large, λ_1 becomes large according to (39). The optimization is then dominated by

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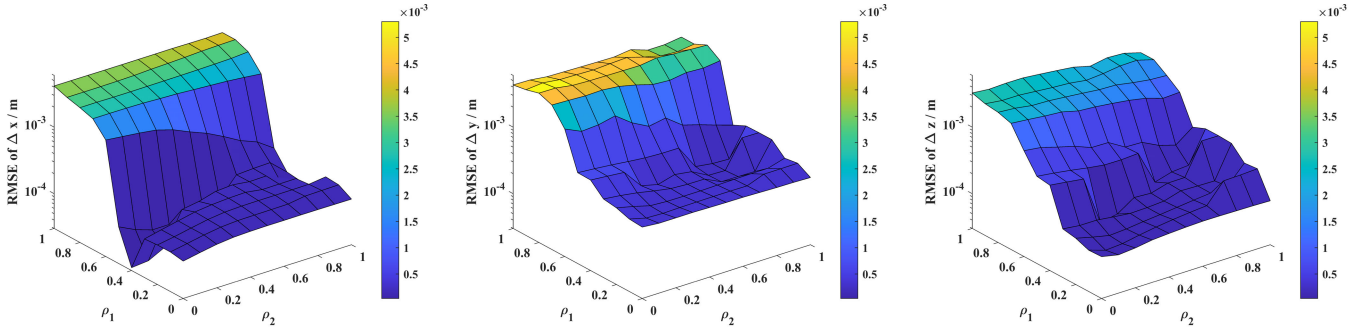


Fig. 11. RMSE of the reconstructed trajectory as a function of the dimensionless weights ρ_1 and ρ_2. (a) Cross-track direction. (b) Along-track direction. (c) Vertical direction.

the corresponding regularization component, which can induce overly aggressive updates in the range-related parameters and lead to overcorrection. Good performance is obtained when ρ_1 is kept moderate. For $\rho_1 < 0.5$, submillimeter reconstruction accuracy is consistently achieved in the along-track, cross-track, and vertical directions. As ρ_1 increases, the RMSE gradually increases. When $\rho_1 \leq 1$, the reconstruction accuracy remains at the millimeter level, which is still satisfactory. Overall, the method is not overly sensitive to the regularization setting within $[0, 1]$. A preferable choice is $\rho_1 < 0.5$ when the range defocus is weak.

B. Results on Measured Data

The measured data were acquired by a Ku-band stripmap MWP-SAR system, provided by the Institute of Aerospace Information Innovation, Chinese Academy of Sciences [4]. The system parameters are consistent with those used in the simulations, as summarized in Table II.

Using the proposed full-trajectory reconstruction method, a trajectory extending over 1880 m is reconstructed. Based on this trajectory, the echo data are processed with TDAs to obtain a UHR-SAR image covering an area of 2×1 km, as illustrated in Fig. 12. Three subscenes are selected for a more detailed presentation of the imaging results. Moreover, the optical image is also provided as a visual reference.

In particular, the imaging results of the building in Scenario 1, a corner reflector, an aircraft, and a square manhole cover in Scenario 2, as well as the round manhole cover in Scenario 3, are highlighted. The compared processing schemes are identical to those used in the simulations, and the corresponding results are presented in Fig. 13. The results confirm the effectiveness of the proposed method for the considered UHR large-scene imaging scenario. For instance, in the case of the building in Scenario 1, the image formed without autofocus exhibits terrain defocusing and an artificially continuous rooftop. The building remains poorly focused after conventional BP autofocus because this scheme only compensates spatial invariant phase errors. Although the MWP-SAR imaging method in [16] improves the focusing to some extent, its performance is limited by the large scene size and the result exhibits noticeable energy leakage on the right side of the rooftop. With the proposed method, the terrain details are well resolved, no observable energy leakage appears around the rooftop, and a distinct gap is visible along the center of the rooftop. For the aircraft in Scenario 2, the image formed using the trajectory reconstructed by the proposed method clearly resolves structural features such as the fuselage, wings, nose, and tail, with clear

outlines and visible internal skeletal structures. Similar observations can be made for both the square and round manhole covers, especially for the square manhole cover, where the frame edges are sharply defined and readily distinguishable. It should be noted that, because of the small size of the manhole cover, conventional BP autofocus produces an almost point-like response, thereby failing to preserve its actual geometric structure. This indicates that, for such small structured targets, sharpness-related metrics should be interpreted together with the displayed target structure.

To further evaluate the imaging performance of the proposed method, quantitative evaluations are conducted on the imaging results obtained by various schemes, using IE, coefficient of variation (CV) contrast, and image gradient sharpness (IS) as metrics [51]. Specifically, the term image contrast (IC) in this article refers exclusively to CV contrast. Given the complex backgrounds and strong scattering variability inherent in SAR imagery, these metrics offer complementary insights into image quality. Specifically, IE quantifies the structural complexity and information content of the reconstructed images, where lower entropy values generally indicate better focusing and reduced randomness. Compared to traditional measures such as standard deviation contrast, CV contrast provides improved normalization and robustness against global intensity variations. Additionally, IS effectively captures edge clarity and target delineation, serving as a reliable indicator of focusing quality while mitigating speckle noise effects. Quantitative comparison results are summarized in Table IV. The proposed method shows consistently favorable performance across the three metrics. Although the IE and IC values of conventional BP autofocus are better than those of the proposed approach in the manhole cover scenario, the corresponding image shows that conventional BP autofocus produces a nearly point-like response that is inconsistent with the actual geometric structure of the target. This indicates that quantitative metrics should be

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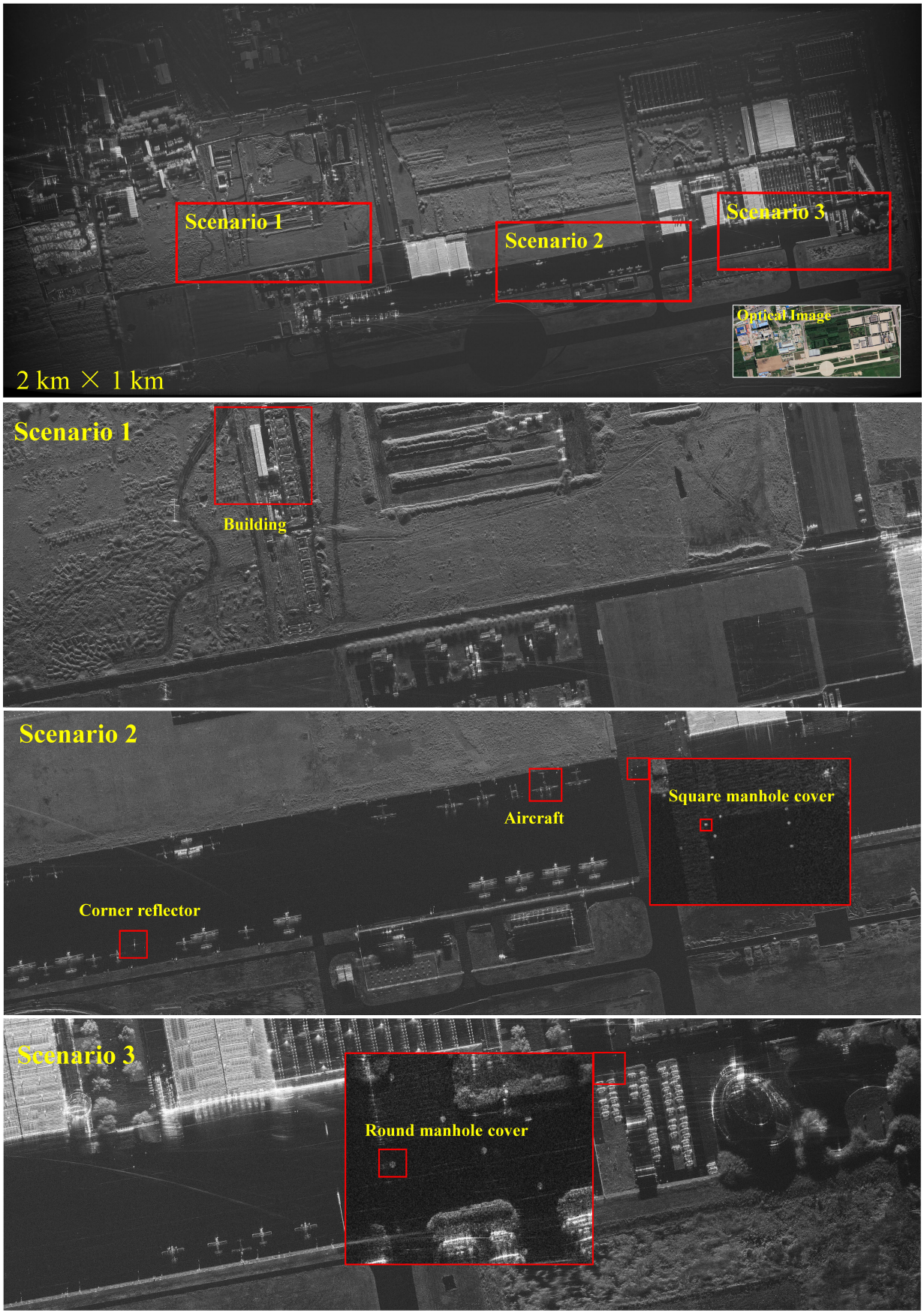


Fig. 12. Imaging result obtained by the proposed method for a 2 × 1 km scene, the corresponding optical reference image, and the enlarged views of three selected subscenes.

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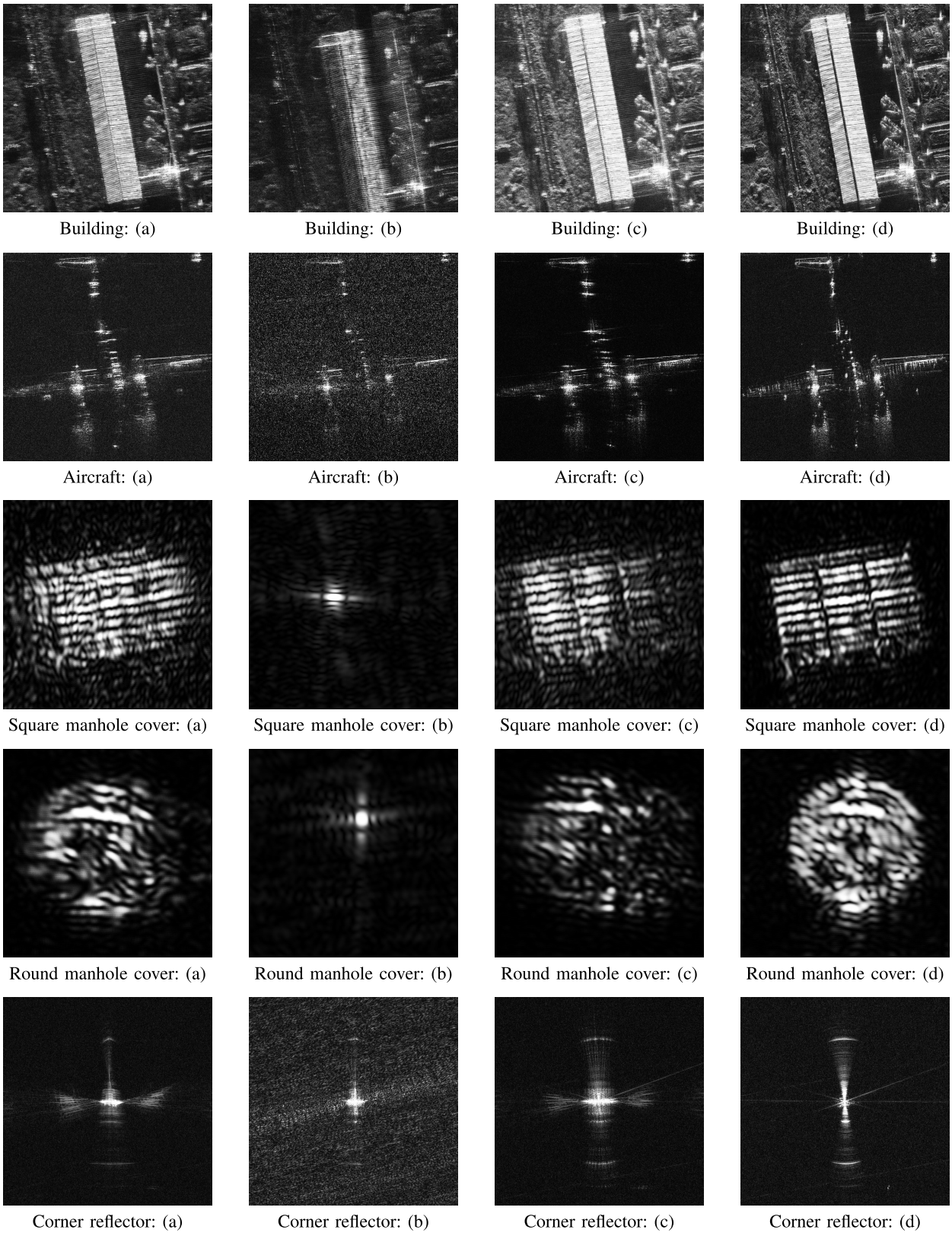


Fig. 13. Comparison of imaging results obtained by different processing schemes for five representative targets. (a) Without autofocus, (b) with conventional BP autofocus [36], (c) MWP-SAR imaging method in [16], and (d) proposed method.

interpreted together with the imaging results to enable a more objective and accurate assessment of image quality.

The imaging results of the corner reflector obtained by different processing schemes are also illustrated in Fig. 13, with the corresponding range and azimuth profiles presented in Fig. 14. The associated IRW, PSLR, and ISLR metrics are summarized in Table V. The theoretical azimuth resolution is 0.019 m. The image formed using the trajectory

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TABLE IV
PERFORMANCE COMPARISON OF IMAGING QUALITY FOR DIFFERENT TARGETS OBTAINED BY DIFFERENT PROCESSING SCHEMES

Scheme	Building			Aircraft			Square manhole cover			Round manhole cover			Corner reflector		
	IE	IC	IS	IE	IC	IS	IE	IC	IS	IE	IC	IS	IE	IC	IS
Without autofocus	13.86	1.610	2.253	13.40	0.8846	2.229	11.70	0.7490	1.377	11.50	0.8914	1.148	9.383	2.877	2.103
Conventional BP autofocus	13.98	1.378	2.21	13.83	0.6558	2.222	8.297	2.097	1.168	9.394	1.624	0.8170	11.00	1.403	1.829
MWP-SAR method in [16]	13.66	1.750	2.259	12.95	1.060	2.188	11.51	0.8511	1.314	11.54	0.8505	1.184	8.632	3.828	2.094
Proposed method	13.50	1.851	2.268	12.41	1.176	2.191	11.48	0.9091	1.358	11.25	1.090	1.385	5.726	4.759	2.107

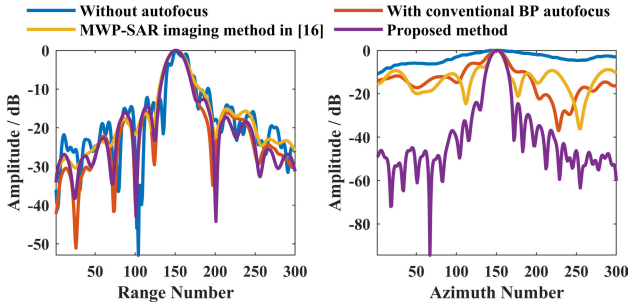


Fig. 14. Range and azimuth profiles for corner reflector obtained using different processing schemes.

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT PROCESSING SCHEMES FOR CORNER REFLECTOR IN TERMS OF RANGE AND AZIMUTH METRICS

Scheme	Range profile			Azimuth profile		
	IRW (m)	PSLR (dB)	ISLR (dB)	IRW (m)	PSLR (dB)	ISLR (dB)
Without autofocus	0.025	-10.03	-6.030	0.148	-7.779	-3.034
Conventional BP autofocus	0.030	-16.55	-10.61	0.028	-10.16	-1.632
MWP-SAR method in [16]	0.032	-14.84	-8.534	0.020	-12.87	-0.321
Proposed method	0.029	-13.66	-9.752	0.020	-24.54	-20.38

reconstructed by the proposed method achieves a resolution of 0.029 × 0.020 m (ground range × azimuth), which is very close to the theoretical azimuth resolution, while also providing satisfactory PSLR and ISLR values. The range profiles obtained by the other schemes still exhibit acceptable focusing quality. However, the azimuth profiles show clear differences. The image formed using only the recorded trajectory shows an excessively large IRW, while the azimuth profiles produced by the other two schemes suffer from significantly high ISLR values, making their azimuth resolution unreliable.

V. CONCLUSION

In this work, a full-trajectory reconstruction method has been developed for UHR-SAR large-scene imaging. By improving the range history model and adopting a dimensionality reduction-based optimization, the proposed method jointly estimates the residual motion errors and dominant target positions, thereby enabling reconstruction of a full-trajectory capable of covering the entire large scene. Simulation and experimental results demonstrate that even when the insufficient precision of the equipped GPS/INS causes severe 2-D defocusing in the initial imaging results, utilizing the trajectory reconstructed by the proposed method still yields well-focused UHR images with centimeter-level resolution. Furthermore, the well-focused 2 × 1 km UHR image obtained by applying the proposed method to airborne experimental data confirms the effectiveness of this approach for UHR-SAR stripmap large-scene applications.

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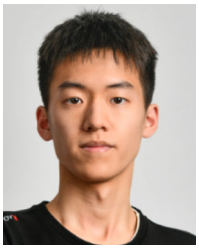
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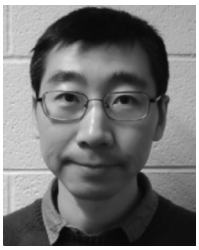
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